AOU
الجامعة العربية المفتوحة
Arab Open University

Faculty of Business Studies

Fin340 and FIN340-F

Course Title: ASSET PRICING AND CORPORATE FINANCE

Version C

Mid-term Examination

2024-2025 / Spring

9-April-2025

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ber of Exam Pages: 7 g this cover sheet)	Time Allowed:1.5 hours

ructions:

IVE (5) minutes to read all instructions carefully before attempting to write.
phones must be in "SWITCHED OFF" mode throughout the examination.
a CLOSED book examination. NO documents of any kind are allowed.
UST answer the questions using only the answer booklets provided
UST return the examination paper together with your answer booklets.
T write your answers in pencil except graphs/sketches
s are NOT allowed to use any type of dictionary or mobile phones during the exam
s are allowed to use *non-programmable* calculators only (if needed).
m of cheating will NOT be tolerated and will subject you to AOU Cheating Policy.

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Solve All Questions

- **Question 1: (30 marks)**

Wayne Corp. had equity of $375,000 at the beginning of the year. At the end of the year, the company had total assets of $300,000. During the year, the company sold no new equity. Net income for the year was $80,000 and dividends were $50,000. What is the sustainable growth rate?

- **Question 2: (30 marks)**

Lassiter Industries has annual sales of $220,000 with 10,000 shares of stock outstanding. The firm has a profit margin of 6 percent and a price-sales ratio of 1.20. What is the firm's price-earnings ratio? Interpret the company's price-earnings ratio

- **Question 3: (25 marks)**

Billings, Inc. has net income of $161,000, a profit margin of 7.6 percent, and an accounts receivable balance of $127,100. Assume that 66 percent of sales are on credit. What is the days' sales in receivables? Interpret your results

- **Question 4: (15 marks)**

What concerns might a loan officer have when loaning funds to a sole proprietorship that he or she might not have when loaning funds to a corporation?

End of Questions

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Formulas

Cash Flow Summary

I. The cash flow identity

Cash flow from assets = Cash flow to creditors (bondholders)
+ stockholders (owners)

II. Cash flow from assets

Cash flow from assets = Operating cash flow
− Net capital spending
− Change in net working capital (NWC)

where:

Operating cash flow = Earnings before interest and taxes (EBIT)
+ Depreciation − Taxes

Net capital spending = Ending net fixed assets − Beginning net fixed assets
+ Depreciation

Change in NWC = Ending NWC − Beginning NWC

III. Cash flow to creditors (bondholders)

Cash flow to creditors = Interest paid − Net new borrowing

IV. Cash flow to stockholders (owners)

Cash flow to stockholders = Dividends paid − Net new equity raised

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Common Financial Ratios

I. Short-term solvency, or liquidity, ratios

$$\text{Current ratio} = \frac{\text{Current assets}}{\text{Current liabilities}}$$

$$\text{Quick ratio} = \frac{\text{Current assets} - \text{Inventory}}{\text{Current liabilities}}$$

$$\text{Cash ratio} = \frac{\text{Cash}}{\text{Current liabilities}}$$

$$\text{Net working capital to total assets} = \frac{\text{Net working capital}}{\text{Total assets}}$$

$$\text{Interval measure} = \frac{\text{Current assets}}{\text{Average daily operating costs}}$$

II. Long-term solvency, or financial leverage, ratios

$$\text{Total debt ratio} = \frac{\text{Total assets} - \text{Total equity}}{\text{Total assets}}$$

$$\text{Debt-equity ratio} = \text{Total debt/Total equity}$$

$$\text{Equity multiplier} = \text{Total assets/Total equity}$$

$$\text{Long-term debt ratio} = \frac{\text{Long-term debt}}{\text{Long-term debt} + \text{Total equity}}$$

$$\text{Times interest earned ratio} = \frac{\text{EBIT}}{\text{Interest}}$$

$$\text{Cash coverage ratio} = \frac{\text{EBIT} + \text{Depreciation}}{\text{Interest}}$$

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III. Asset management, or turnover, ratios

$$\text{Inventory turnover} = \frac{\text{Cost of goods sold}}{\text{Inventory}}$$

$$\text{Days' sales in inventory} = \frac{\text{365 days}}{\text{Inventory turnover}}$$

$$\text{Receivables turnover} = \frac{\text{Sales}}{\text{Accounts receivable}}$$

$$\text{Days' sales in receivables} = \frac{\text{365 days}}{\text{Receivables turnover}}$$

$$\text{NWC turnover} = \frac{\text{Sales}}{\text{NWC}}$$

$$\text{Fixed asset turnover} = \frac{\text{Sales}}{\text{Net fixed assets}}$$

$$\text{Total asset turnover} = \frac{\text{Sales}}{\text{Total assets}}$$

IV. Profitability ratios

$$\text{Profit margin} = \frac{\text{Net income}}{\text{Sales}}$$

$$\text{Return on assets (ROA)} = \frac{\text{Net income}}{\text{Total assets}}$$

$$\text{Return on equity (ROE)} = \frac{\text{Net income}}{\text{Total equity}}$$

$$\text{ROE} = \frac{\text{Net income}}{\text{Sales}} \times \frac{\text{Sales}}{\text{Assets}} \times \frac{\text{Assets*}}{\text{Equity}}$$

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V. Market value ratios

$$\text{Price-earnings ratio} = \frac{\text{Price per share}}{\text{Earnings per share}}$$

$$\text{PEG ratio} = \frac{\text{Price-earnings ratio}}{\text{Earnings growth rate (\%)}}$$

$$\text{Price-sales ratio} = \frac{\text{Price per share}}{\text{Sales per share}}$$

$$\text{Market-to-book-ratio} = \frac{\text{Market value per share}}{\text{Book value per share}}$$

$$\text{Tobin's Q ratio} = \frac{\text{Market value of assets}}{\text{Replacement cost of assets}}$$

$$\text{Enterprise value-EBITDA ratio} = \frac{\text{Enterprise value}}{\text{EBITDA}}$$

ROE= Profit Margin X Total Asset Turnover X Equity Multiplier

Summary of Internal and Sustainable Growth Rates

I. Internal Growth Rate

$$\text{Internal growth rate} = \frac{\text{ROA} \times b}{1 - \text{ROA} \times b}$$

where:

ROA = Return on assets= Net income/Total assets
b = Plowback (retention) ratio
= Addition to retained earnings/Net income

The internal growth rate is the maximum growth rate that can be achieved with no external financing of any kind.

II. Sustainable Growth Rate

$$\text{Sustainable growth rate} = \frac{\text{ROE} \times b}{1 - \text{ROE} \times b}$$

where:

ROE = Return on equity= Net income/Total equity
b = Plowback (retention) ratio
= Addition to retained earnings/Net income

The sustainable growth rate is the maximum growth rate that can be achieved with no external equity financing while maintaining a constant debt-equity ratio

Dividend payout ratio = Cash dividends/Net income

Retention ratio= 1- the dividend payout ratio

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$$EPS = \frac{\text{Net income}}{\text{Shares outstanding}} =$$

Net Working Capital= Current Asset- Current Liabilities

Present value: PV = $\mathbf{FV/ (1 + r)^t}$

Future Value: FV = $\mathbf{PV*(1 + r)}$

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